PHILIPS

DICOM Information Model

Agenda

- Evolution of Information
- Building Blocks of DICOM
  - Attribute
  - Information Object Module
  - Information Entity
  - Information Object Definition
- SOP Class & SOP Instance

Patient

Patient Name

Patient Birth-date

Patient Sex

Patient Address

Patient Phone Number

Patient Allergy

Patient History

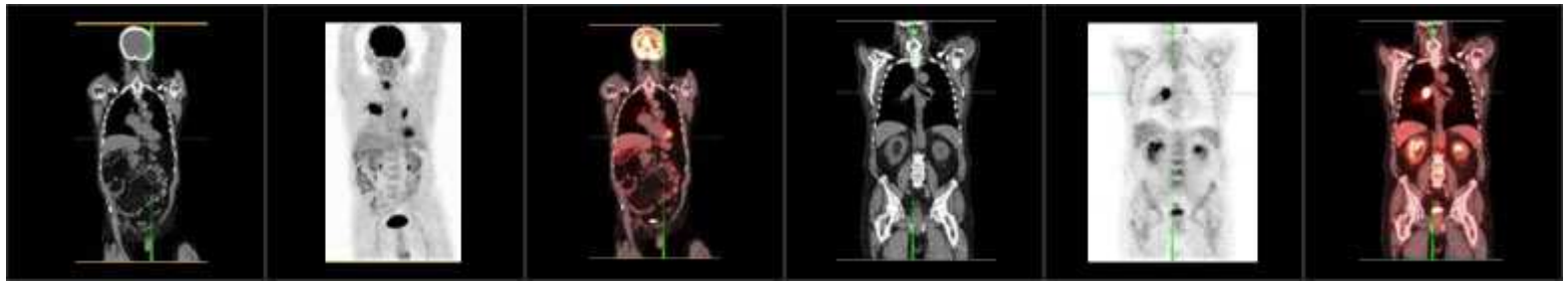
Study

Physician’s name
Referring Physician
Imaging Request
Study Date/Time
Scheduled Scan
Patient Weight

Scan

Modality
Scan Type
Manufacturer
Device Serial Number
Body Parts Examined
Scan Date/Time

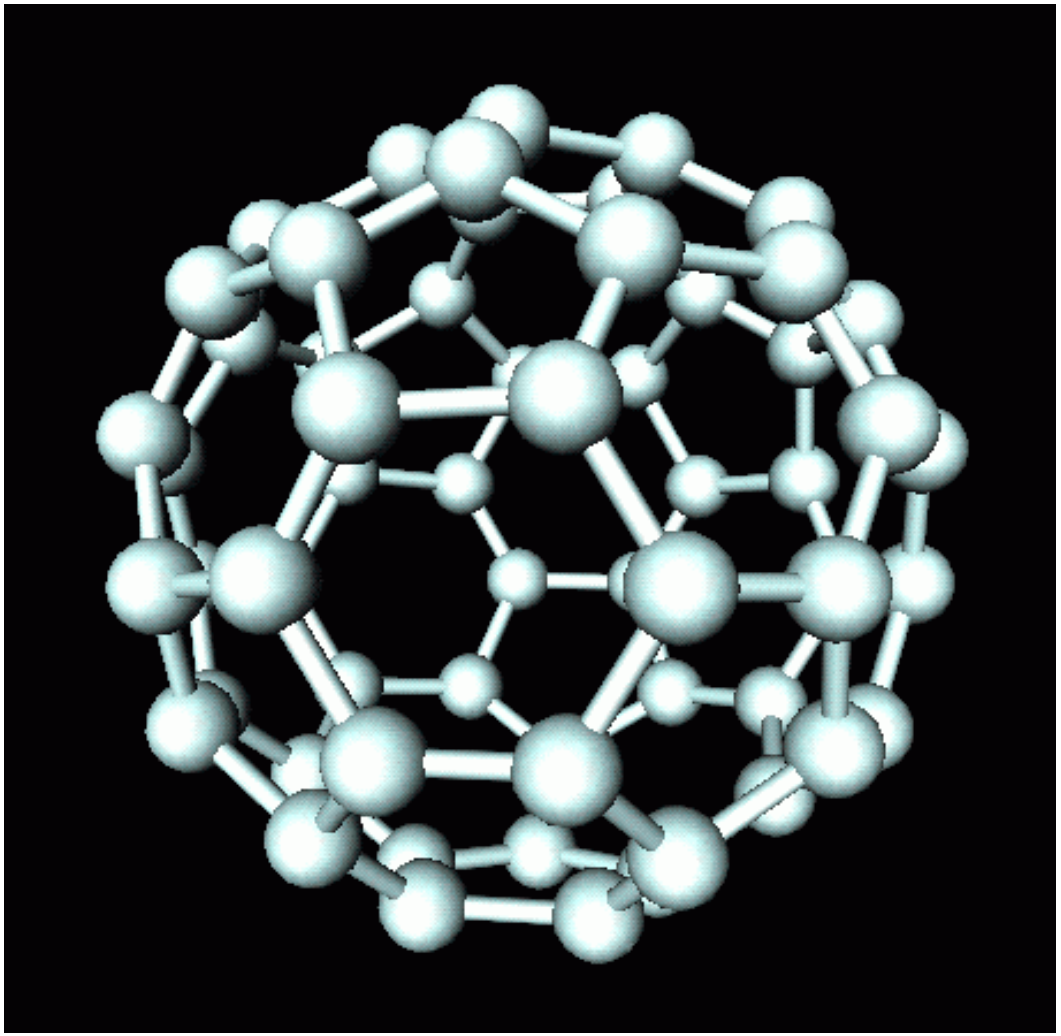
Imaging



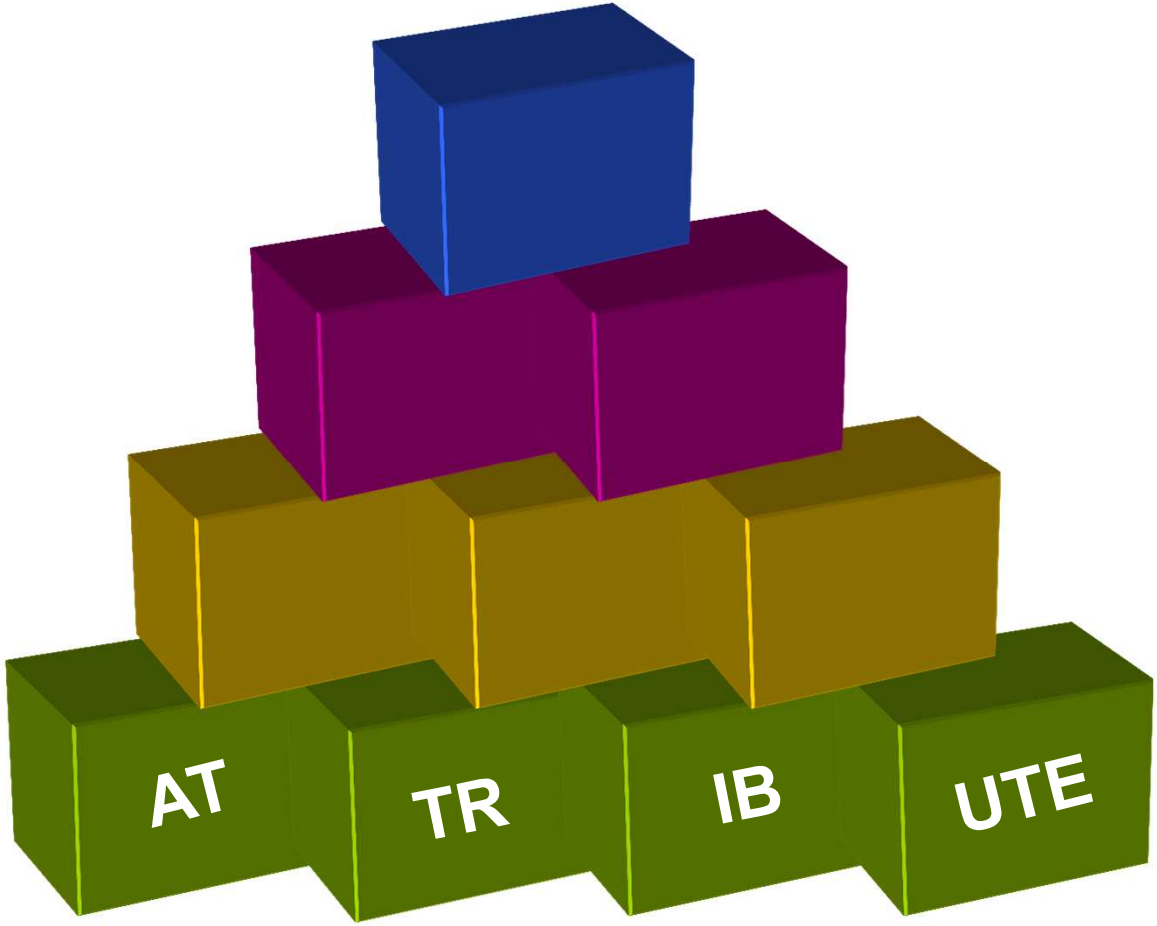
Patient Orientation Information
Contrast Agent Information
Image Pixel Information
Pixel Data
Overlays and Annotations

Random Data

Structured Representation

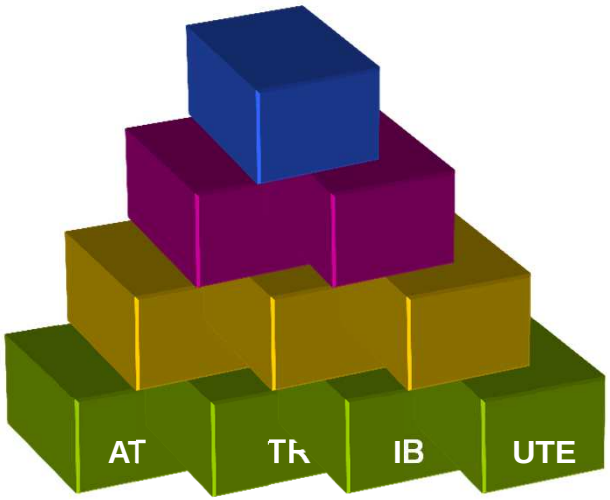


Building Blocks of DICOM



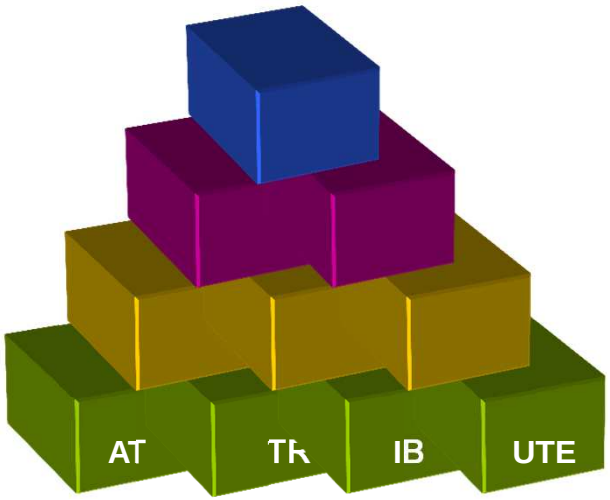
Attribute

- Anatomy of an Attribute
  - Name
  - Tag
  - Value Representation
  - Value Multiplicity
  - Value



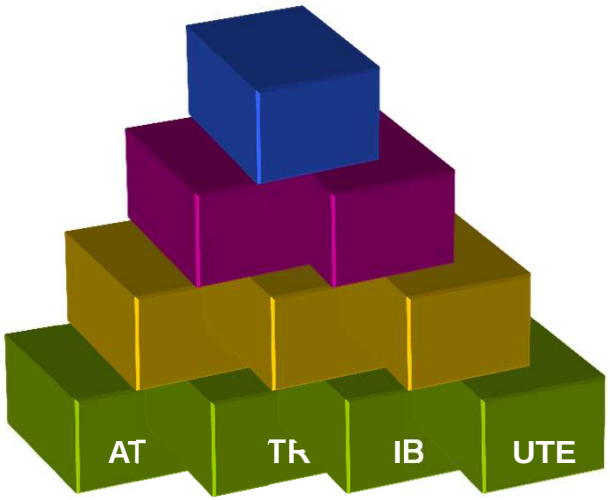
Attribute

- Tag
  - Hexadecimal Number
    - Group gggg
    - Element eeee
  - Group
    - Even – DICOM Defined
    - Odd – Vendor Defined
  - Example
    - 0010 0010 – Patient Name
    - 0010 0020 – Patient ID
    - 0010 0030 – Patient Birth Date



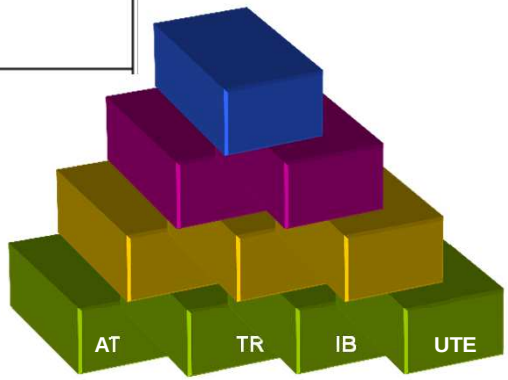
Attribute

- Value Representation (VR)
  - Data Type and the Format
  - Length
  - Permissible Characters
  - Categories
    - Mathemetical – FL, SS, DS etc
    - Character String – CS, ST etc
    - Special Characteristics – DA, DT etc
    - Other – OB, OW and SQ



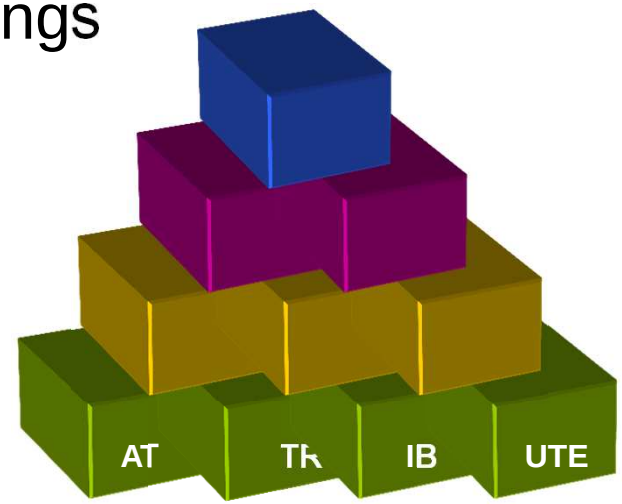
Attribute: Value Representation

VR Name	Definition	Character Repertoire	Length of Value
CS Code String	A string of characters with leading or trailing spaces (20H) being non-significant.	Uppercase characters, "0"-"9", the SPACE character, and underscore "_", of the Default Character Repertoire	16 bytes maximum
FL Floating Point Single	Single precision binary floating point number represented in IEEE 754:1985 32-bit Floating Point Number Format.	not applicable	4 bytes fixed
IS Integer String	A string of characters representing an Integer in base-10 (decimal), shall contain only the characters 0 - 9, with an optional leading "+" or "-". It may be padded with leading and/or trailing spaces. Embedded spaces are not allowed. The integer, n, represented shall be in the range: $-2^{31} <= n <= (2^{31} - 1)$.	"0"-"9", "+", "-" of Default Character Repertoire	12 bytes maximum

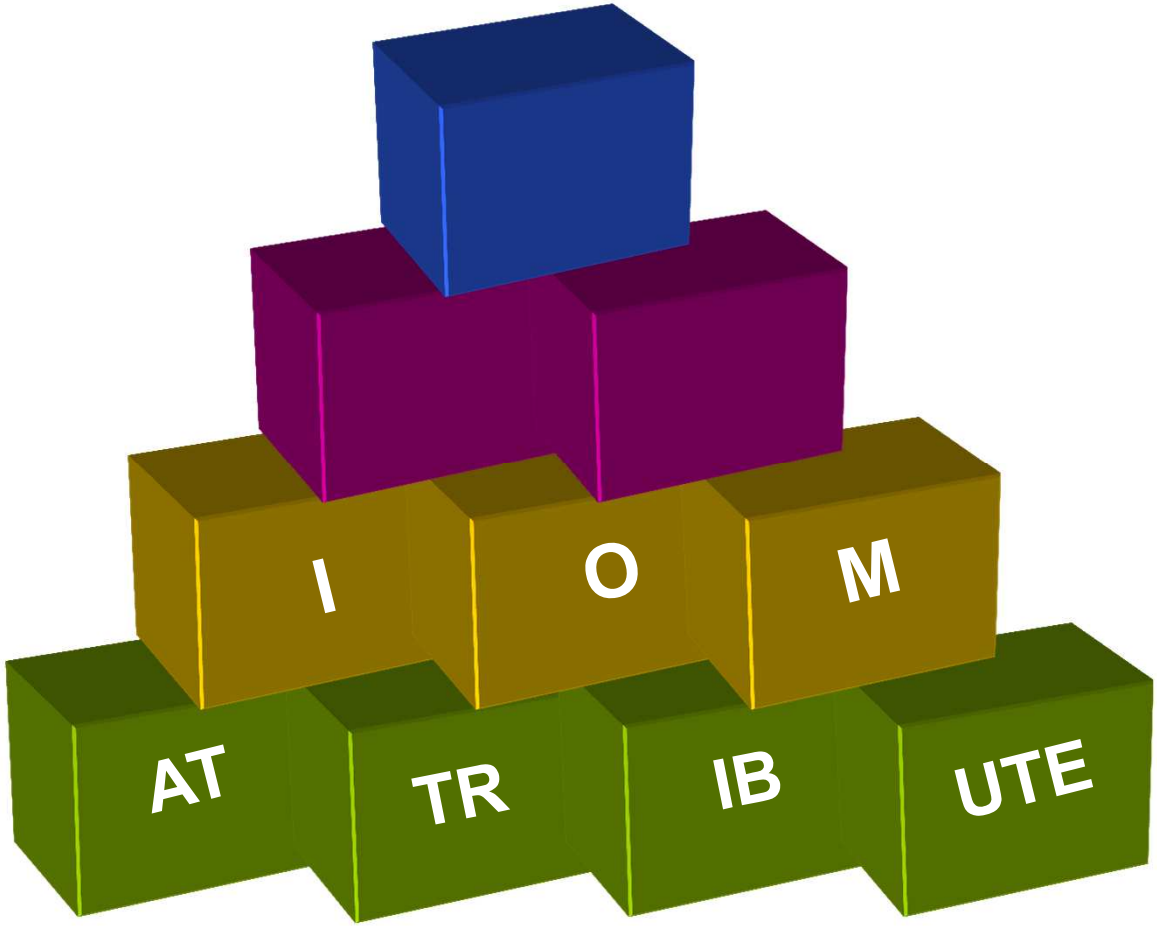


Attribute

- Value Multiplicity
  - Number of values for an Attribute
    - Example:
      - VM = 1
      - VM = 1-n
  - “\” is the Delimiter for Character Strings
  - No delimiter for Fixed Length VR

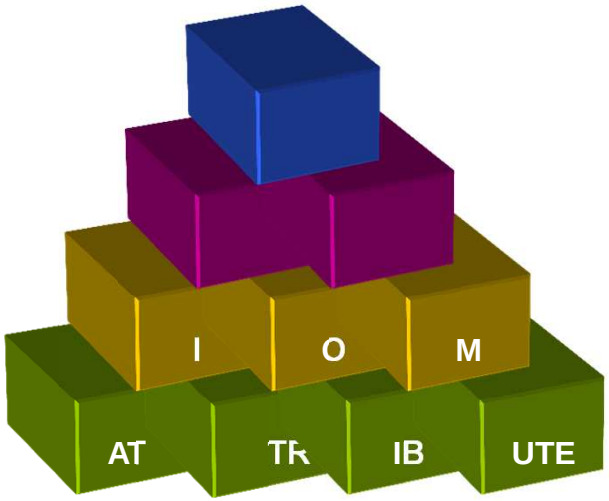


Building Blocks of DICOM



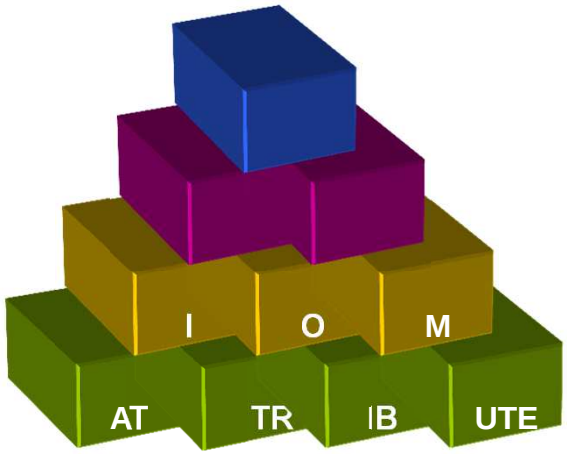
Information Object Module (IOM)

- Logical Grouping of Related Attributes
- Attribute Description
- Mandatoriness of Attribute
  - 1 Required and non null
  - 1C Required based on a condition
  - 2 Required can be null
  - 2C Required based on a condition
  - 3 Optional

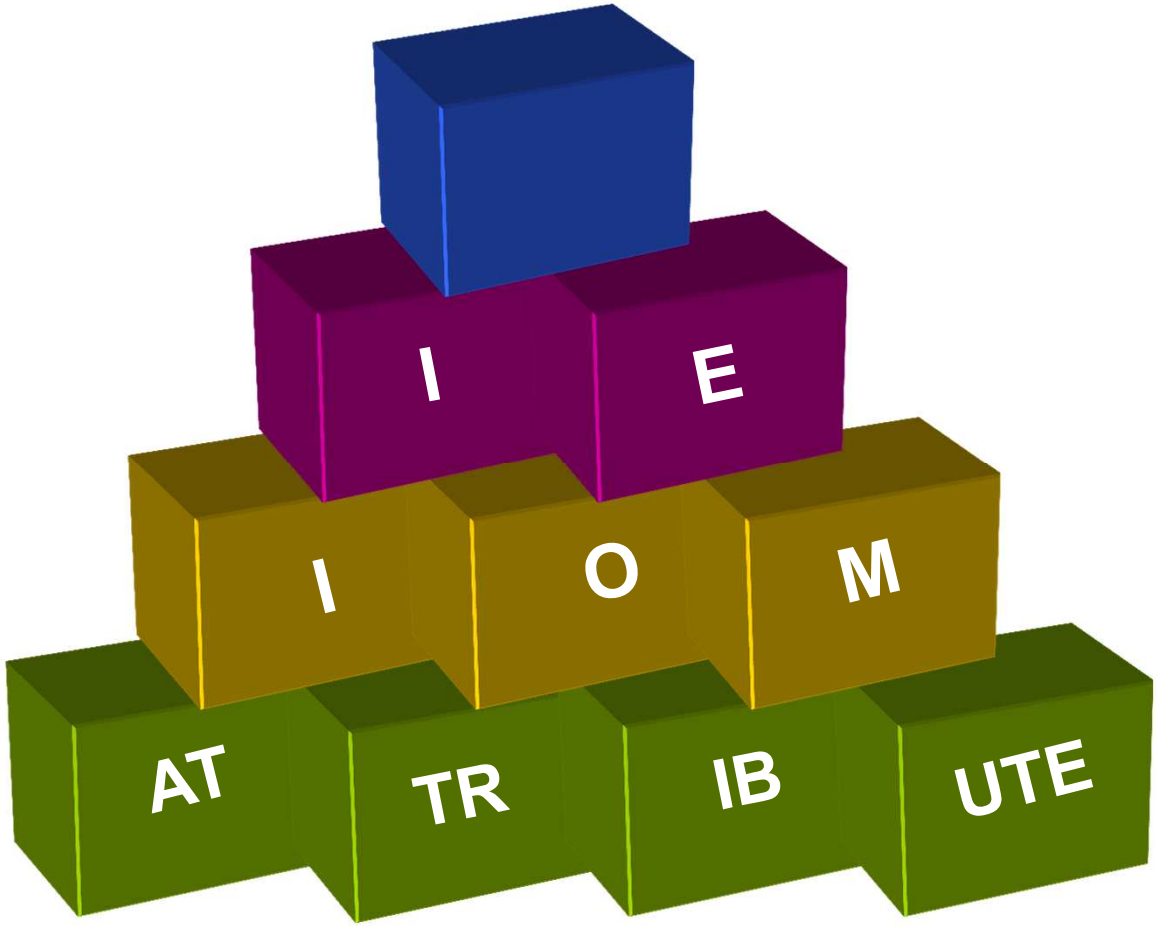


Information Object Module
Series Module

Attribute Name	Tag	Type	Attribute Description
Modality	(0008,0060)	1	Type of equipment that originally acquired the data used to create the images in this Series. See C.7.3.1.1.1 for Defined Terms.
Series Instance UID	(0020,000E)	1	Unique identifier of the Series.
Series Number	(0020,0011)	2	A number that identifies this Series.
Laterality	(0020,0060)	2C	Laterality of (paired) body part examined. Required if the body part examined is a paired structure and Image Laterality (0020,0062) or Frame Laterality
Series Date	(0008,0021)	3	Date the Series started.
Series Time	(0008,0031)	3	Time the Series started.
Performing Physicians' Name	(0008,1050)	3	Name of the physician(s) administering the Series.

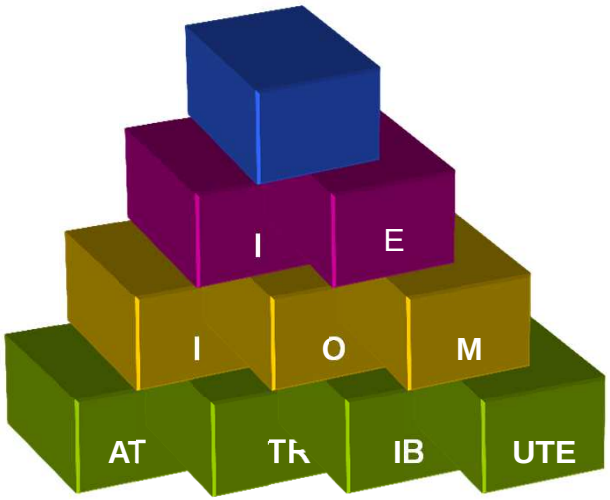


Building Blocks of DICOM



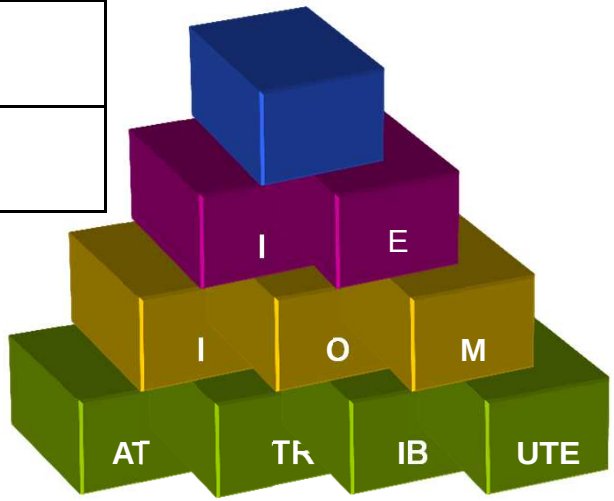
Information Entity (IE)

- Collection of one or more related Modules
- Represents a Single Real-World Information
  e.g: Patient, Study, Series, Image
- Usage of the Module
  - M Mandatory
  - U User Optional

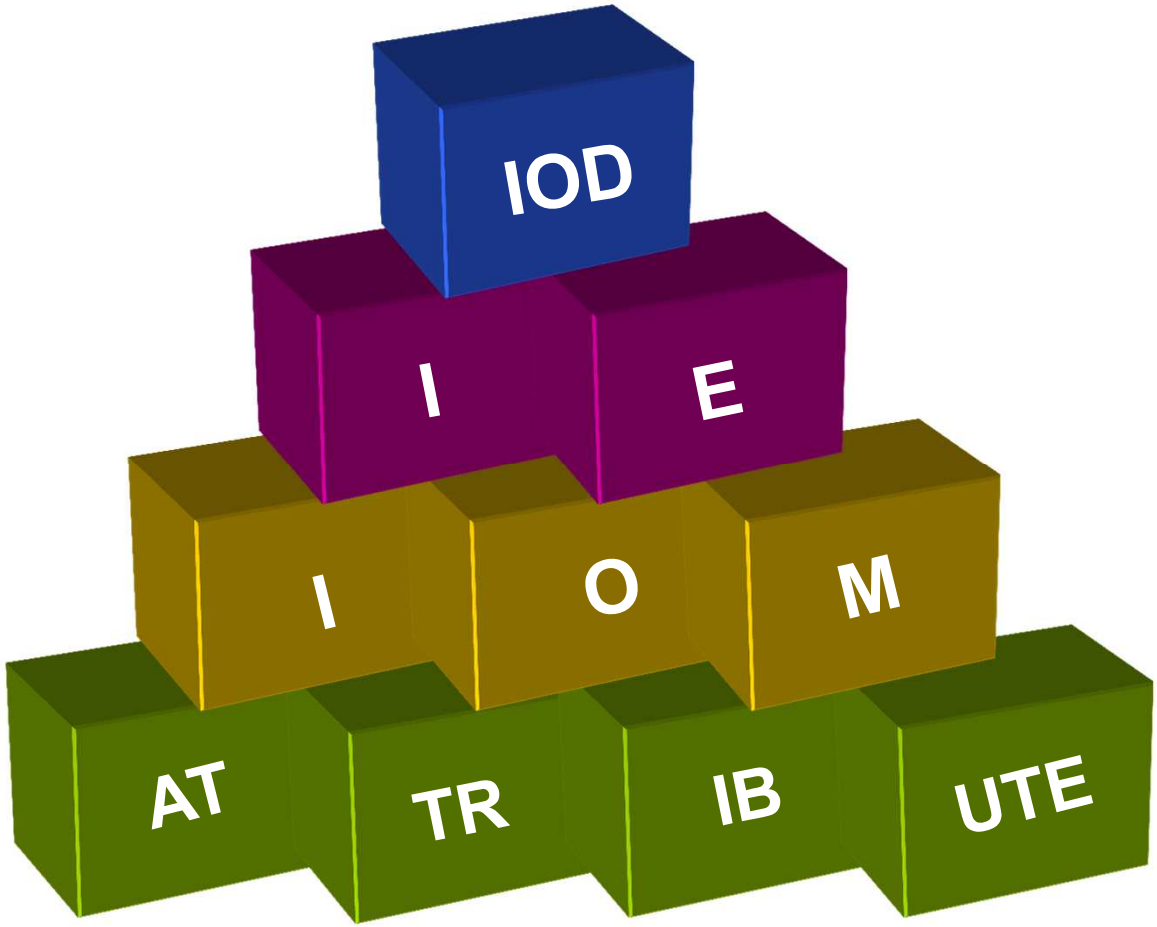


Information Entity (IE) Example

IE	Module	Usage
Patient	Patient	M
	Clinical Trial Subject	U
Study	General Study	M
	Patient Study	U
	Clinical Trail Study	U



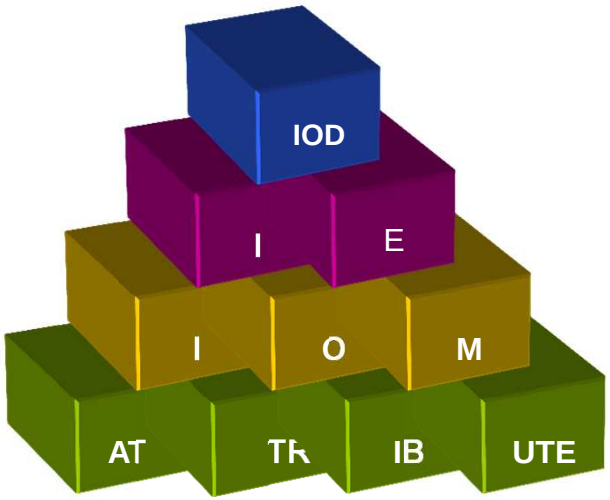
Building Blocks of DICOM



Information Object Definition (IOD)

- A Collection of one or more IE's
- Representing a Real World Entity

  e.g:
  - MR Image
  - XA Image
  - Printing Images



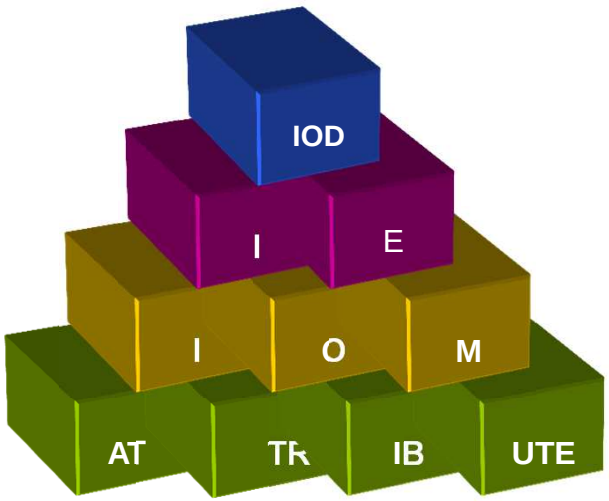
Information Object Definition (IOD)

- Composite IOD
  - Contains one or more IEs
  - Persistent Objects
  - Used for Complex Transactions

    e.g: Storage, Query/Retrieve

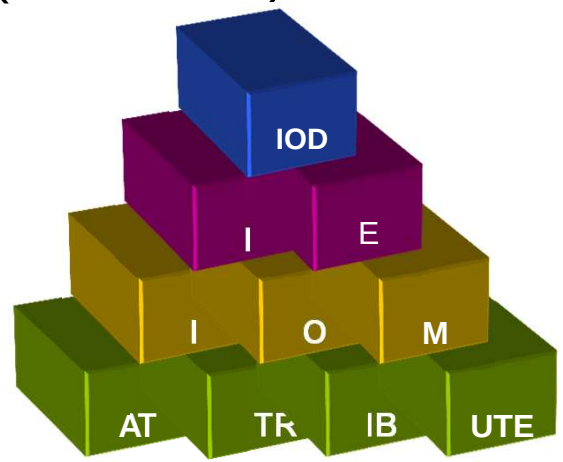
- Normalized IOD
  - Contains only one IE
  - Temporary Objects
  - Used for Management Transactions

    e.g: Print



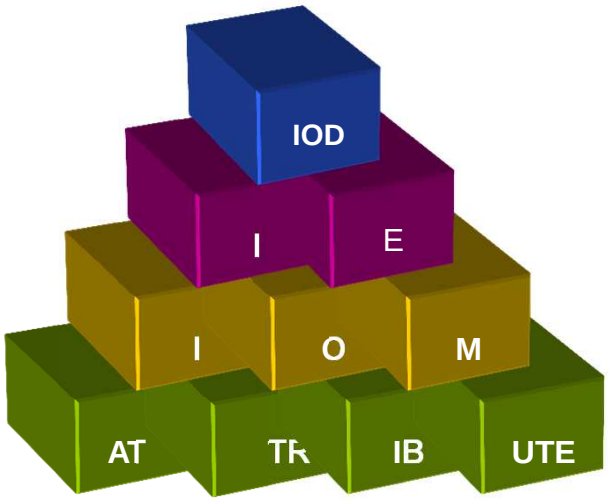
Information Object Definition (IOD)

- IODs are Identified by SOP Class
- SOP Class
  - SOP Class = Service + Object (IOD)
  - Defines the Type of information Stored along with Service associated (e.g: MR Image Storage, Patient Query etc)
  - Identified by an Unique Identifier called SOPClassUID
  - At least one SOP Class for each Modality (MR/CT/US)

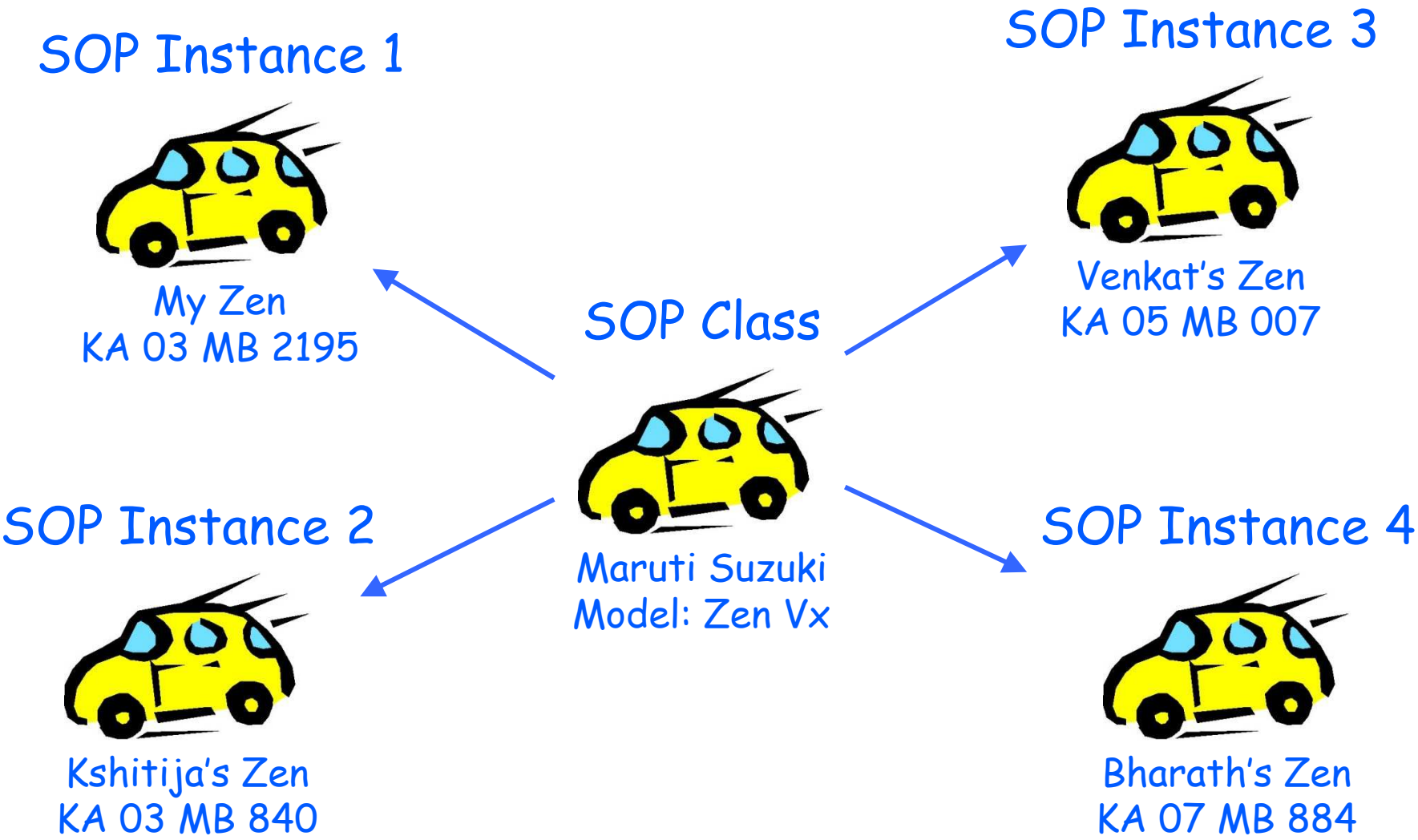


SOP Instance

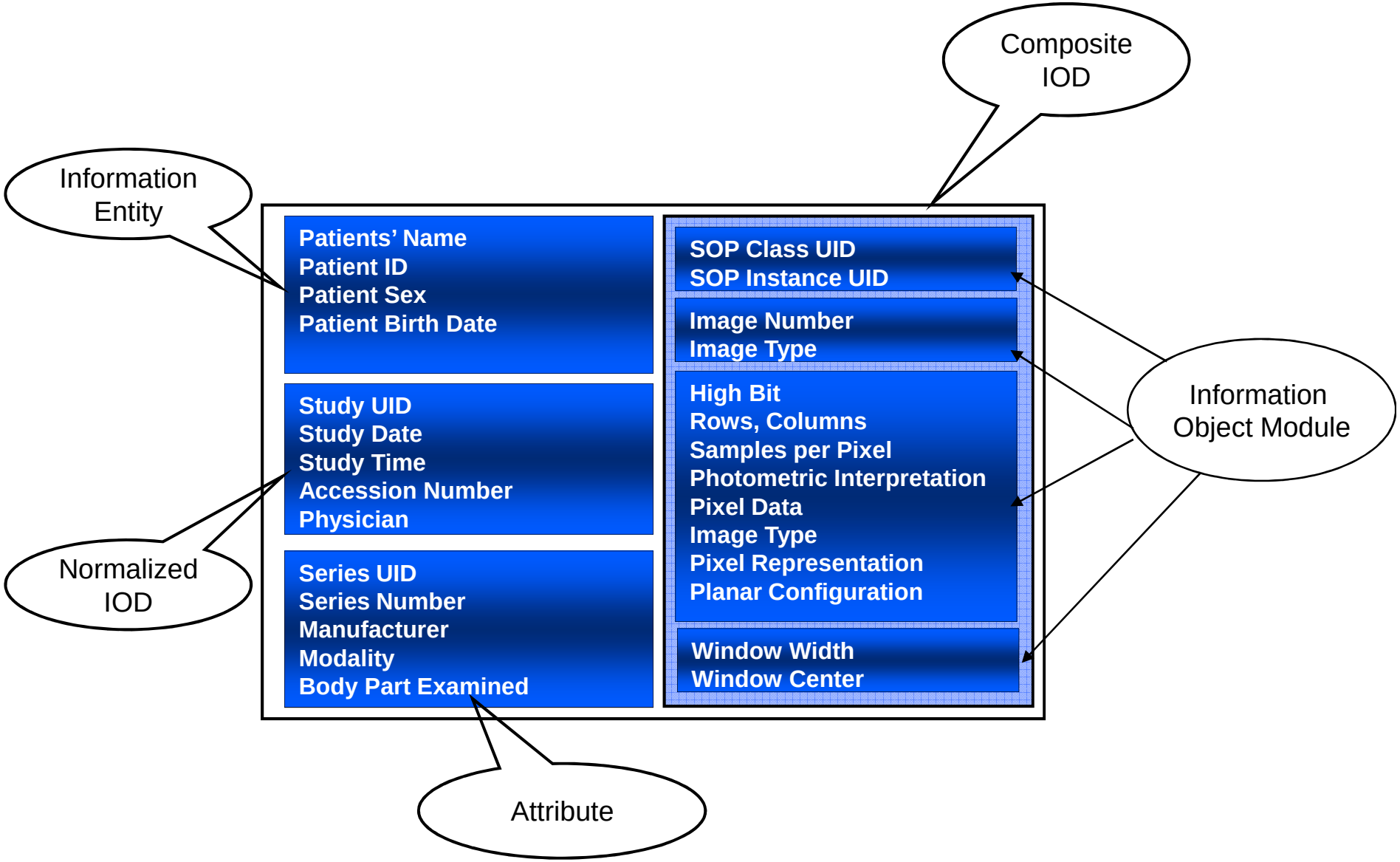
- Instance of a SOP Class
- Similar to OO Concepts Class/Instance
- Each Instance is identified by an Unique Identifier called SOP Instance UID



Information Object Definition (IOD)



DICOM IOD


Composite IOD
Information Entity
Patients' Name
Patient ID
Patient Sex
Patient Birth Date
Study UID
Study Date
Study Time
Accession Number
Physician
Series UID
Series Number
Manufacturer
Modality
Body Part Examined
SOP Class UID
SOP Instance UID
Image Number
Image Type
High Bit
Rows, Columns
Samples per Pixel
Photometric Interpretation
Pixel Data
Image Type
Pixel Representation
Planar Configuration
Window Width
Window Center
Information Object Module
Normalized IOD
Attribute

Computed Radiography Image IOD

IE	Module
Patient	Patient
Study	General Study
	Patient Study
Series	General Series
	CR Series
Equipment	General Equipment
Image	General Image
	Image Pixel
	Contrast/bolus
	CR Image
	Overlay Plane
	Curve
	Modality LUT
	VOILUT
	SOP Common

Data Dictionary

CR IOD

IE	Module	Attribute

CT IOD

IE	Module	Attribute

MR IOD

IE	Module	Attribute

XA IOD

IE	Module	Attribute

SC IOD

IE	Module	Attribute

US IOD

IE	Module	Attribute

PET IOD

IE	Module	Attribute

SPECT IOD

IE	Module	Attribute

Exercise

- What Type of Attributes the following tags belong to
  - 0008, 0016
  - 000B, 0016
  - 0019, 0016

- Interpret the following string “Shiva/Bhaskaran\Geetha\Raghu/Patil” What is the VM of this attribute?

PHILIPS

sense and simplicity